

Exercise 1 • SQL Fundamental: SELECT & Filtering

01. SELECT *
FROM employees;

Table employees

id	First name	Last name	department	Salary	hire date	City
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Motoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

02. SELECT DISTINCT department
FROM employees;

Table employees

department
IT
HR
Finance
Marketing

03. SELECT first_name,
last_name

FROM employees

ORDER BY salary DESC;

Table employees

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

04. SELECT id, first_name, last_name, salary

FROM employees

ORDER BY salary DESC

LIMIT 3;

Table employees

id	first_name	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

05. SELECT id, first_name, last_name, department
 FROM employees
 WHERE department = 'IT';

Table employees

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

06. SELECT id, first_name, last_name, department, salary
 FROM employees
 WHERE department = 'Finance' AND salary > 60000;

Table employees

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

07. SELECT id, first_name, last_name, department
 FROM employees
 WHERE department = 'HR' OR 'Marketing';

Table employees

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08. `SELECT id, first_name, last_name, department`
`FROM employees`

`WHERE NOT department = 'IT';`

Table employees

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09. `SELECT id, first_name, last_name, department`
`FROM employees`
`WHERE department IN ('IT', 'HR', 'Finance');`

Table employees

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. `SELECT id, first_name, last_name, department, salary, city`
`FROM employees`
`WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';`

Table employees

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg